



Carrera: *Ingeniería en Sistemas Computacionales*

Asignatura:

Taller de sistemas operativos

Profesor:

Gustavo Moisés Romero González

Integrantes:

RAMIREZ RIVERA JOSIAS DANIEL

VAZQUEZ BARRANCO BRAYAN DAVID

VIELMA SANCHEZ ESMERALDA

Grupo: **6S12**

“SERVIDOR NGINX”

Fecha de entrega:

25/MAYO/2026

#Objetivo General

Implementar un servidor web NGINX compilado desde código fuente junto con PHP-FPM, utilizando comunicación FastCGI mediante Linux, garantizando el funcionamiento correcto del servidor web y su administración mediante SystemD.

#Objetivos Específicos

- Compilar NGINX desde código fuente versión 1.31.x.
- Configurar PHP-FPM versión 8.4.x.
- Implementar comunicación FastCGI.
- Configurar servicios mediante SystemD.
- Habilitar socket UNIX.
- Verificar funcionamiento con phpinfo().

DESARROLLO DE PROYECTO

Se instalaron las librerías necesarias para la compilación de NGINX y PHP desde código fuente.

```
dramirez@10:~ — sudo dnf update -y

[dramirez@10 ~]$ sudo dnf update -y

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for dramirez: 
```

```
dramirez@10:~

tar-2:1.34-9.el9_7.x86_64
tzdata-2026a-1.el9.noarch
util-linux-2.37.4-21.el9_7.x86_64
util-linux-core-2.37.4-21.el9_7.x86_64
util-linux-user-2.37.4-21.el9_7.x86_64
vim-common-2:8.2.2637-23.el9_7.3.x86_64
vim-enhanced-2:8.2.2637-23.el9_7.3.x86_64
vim-filesystem-2:8.2.2637-23.el9_7.3.noarch
vim-minimal-2:8.2.2637-23.el9_7.3.x86_64
webkit2gtk3-2.52.3-0.el9_7.1.x86_64
webkit2gtk3-jsc-2.52.3-0.el9_7.1.x86_64
xorg-x11-server-Xorg-1.20.11-33.el9_7.x86_64
xorg-x11-server-Xwayland-23.2.7-6.el9_7.x86_64
xorg-x11-server-common-1.20.11-33.el9_7.x86_64
Instalado:
grub2-tools-efi-1:2.06-114.el9_7.1.alma.1.x86_64
grub2-tools-extra-1:2.06-114.el9_7.1.alma.1.x86_64
kernel-5.14.0-611.55.1.el9_7.x86_64
kernel-core-5.14.0-611.55.1.el9_7.x86_64
kernel-modules-5.14.0-611.55.1.el9_7.x86_64
kernel-modules-core-5.14.0-611.55.1.el9_7.x86_64
¡Listo!
[dramirez@10 ~]$ sudo dnf groupin
```

```
dramirez@10:~  
grub2-tools-efi-1:2.06-114.el9_7.1.alma.1.x86_64  
grub2-tools-extra-1:2.06-114.el9_7.1.alma.1.x86_64  
kernel-5.14.0-611.55.1.el9_7.x86_64  
kernel-core-5.14.0-611.55.1.el9_7.x86_64  
kernel-modules-5.14.0-611.55.1.el9_7.x86_64  
kernel-modules-core-5.14.0-611.55.1.el9_7.x86_64  
  
;Listo!  
[dramirez@10 ~]$ sudo dnf groupinstall "Development Tools" -y  
[sudo] password for dramirez:  
Última comprobación de caducidad de metadatos hecha hace 0:26:02, el dom 24 may 2026 14:43:20.  
Dependencias resueltas.  
=====
```

Paquete	Arq.	Versión	Repositorio	Tam.
---------	------	---------	-------------	------

```
=====
```

Instalando los paquetes del grupo/módulo:

asciidoc	noarch	9.1.0-3.el9	appstream	237 k
autoconf	noarch	2.69-41.el9	appstream	665 k
automake	noarch	1.16.2-8.el9	appstream	662 k
bison	x86_64	3.7.4-5.el9	appstream	920 k
byacc	x86_64	2.0.20210109-4.el9	appstream	88 k
diffstat	x86_64	1.64-6.el9	appstream	43 k

```
-----
```

```
dramirez@10:~  
redhat-rpm-config-210-1.el9.alma.1.noarch  
rpm-build-4.16.1.3-39.el9.x86_64  
rpm-sign-4.16.1.3-39.el9.x86_64  
rust-srpm-macros-17-4.el9.noarch  
sgml-common-0.6.3-58.el9.noarch  
source-highlight-3.1.9-12.el9.x86_64  
systemtap-5.3-3.el9.x86_64  
systemtap-client-5.3-3.el9.x86_64  
systemtap-devel-5.3-3.el9.x86_64  
systemtap-runtime-5.3-3.el9.x86_64  
tbb-2020.3-9.el9.x86_64  
tzdata-java-2026a-1.el9.noarch  
valgrind-1:3.25.1-3.el9.x86_64  
valgrind-devel-1:3.25.1-3.el9.x86_64  
valgrind-docs-1:3.25.1-3.el9.x86_64  
valgrind-gdb-1:3.25.1-3.el9.x86_64  
valgrind-scripts-1:3.25.1-3.el9.x86_64  
xorg-x11-fonts-IS08859-1-100dpi-7.5-33.el9.noarch  
xz-devel-5.2.5-8.el9_0.x86_64  
zlib-devel-1.2.11-40.el9.x86_64  
zstd-1.5.5-1.el9.x86_64  
  
;Listo!  
[dramirez@10 ~]$
```

```
dramirez@10:~  
valgrind-1:3.25.1-3.el9.x86_64  
valgrind-devel-1:3.25.1-3.el9.x86_64  
valgrind-docs-1:3.25.1-3.el9.x86_64  
valgrind-gdb-1:3.25.1-3.el9.x86_64  
valgrind-scripts-1:3.25.1-3.el9.x86_64  
xorg-x11-fonts-IS08859-1-100dpi-7.5-33.el9.noarch  
xz-devel-5.2.5-8.el9_0.x86_64  
zlib-devel-1.2.11-40.el9.x86_64  
zstd-1.5.5-1.el9.x86_64  
  
¡Listo!  
[dramirez@10 ~]$ sudo dnf install wget tar gcc gcc-c++ make -y  
[sudo] password for dramirez:  
Última comprobación de caducidad de metadatos hecha hace 0:33:04, el dom 24 may 2026 14:43:20.  
El paquete wget-1.21.1-8.el9_4.x86_64 ya está instalado.  
El paquete tar-2:1.34-9.el9_7.x86_64 ya está instalado.  
El paquete gcc-11.5.0-11.el9.alma.1.x86_64 ya está instalado.  
El paquete gcc-c++-11.5.0-11.el9.alma.1.x86_64 ya está instalado.  
El paquete make-1:4.3-8.el9.x86_64 ya está instalado.  
Dependencias resueltas.  
Nada por hacer.  
¡Listo!  
[dramirez@10 ~]$
```

```
dramirez@10:~  
-----  
Total                               294 kB/s | 850 kB    00:02  
Ejecutando verificación de operación  
Verificación de operación exitosa.  
Ejecutando prueba de operaciones  
Prueba de operación exitosa.  
Ejecutando operación  
  Preparando      :                               1/1  
  Instalando      : pcre-utf32-8.44-4.el9.x86_64    1/4  
  Instalando      : pcre-utf16-8.44-4.el9.x86_64    2/4  
  Instalando      : pcre-cpp-8.44-4.el9.x86_64       3/4  
  Instalando      : pcre-devel-8.44-4.el9.x86_64     4/4  
  Ejecutando scriptlet: pcre-devel-8.44-4.el9.x86_64 4/4  
  Verificando      : pcre-cpp-8.44-4.el9.x86_64     1/4  
  Verificando      : pcre-devel-8.44-4.el9.x86_64    2/4  
  Verificando      : pcre-utf16-8.44-4.el9.x86_64    3/4  
  Verificando      : pcre-utf32-8.44-4.el9.x86_64    4/4  
  
Instalado:  
  pcre-cpp-8.44-4.el9.x86_64      pcre-devel-8.44-4.el9.x86_64  
  pcre-utf16-8.44-4.el9.x86_64    pcre-utf32-8.44-4.el9.x86_64  
  
¡Listo!  
[dramirez@10 ~]$
```

```
dramirez@10:~  
Verificación de operación exitosa.  
Ejecutando prueba de operaciones  
Prueba de operación exitosa.  
Ejecutando operación  
Preparando : 1/1  
Instalando : sqlite-3.34.1-9.el9_7.x86_64 1/5  
Instalando : cmake-filesystem-3.26.5-3.el9_7.x86_64 2/5  
Instalando : libxml2-devel-2.9.13-14.el9_7.x86_64 3/5  
Instalando : sqlite-devel-3.34.1-9.el9_7.x86_64 4/5  
Instalando : bzip2-devel-1.0.8-10.el9_5.x86_64 5/5  
Ejecutando scriptlet: bzip2-devel-1.0.8-10.el9_5.x86_64 5/5  
Verificando : bzip2-devel-1.0.8-10.el9_5.x86_64 1/5  
Verificando : cmake-filesystem-3.26.5-3.el9_7.x86_64 2/5  
Verificando : libxml2-devel-2.9.13-14.el9_7.x86_64 3/5  
Verificando : sqlite-3.34.1-9.el9_7.x86_64 4/5  
Verificando : sqlite-devel-3.34.1-9.el9_7.x86_64 5/5  
  
Instalado:  
bzip2-devel-1.0.8-10.el9_5.x86_64 cmake-filesystem-3.26.5-3.el9_7.x86_64  
libxml2-devel-2.9.13-14.el9_7.x86_64 sqlite-3.34.1-9.el9_7.x86_64  
sqlite-devel-3.34.1-9.el9_7.x86_64  
  
¡Listo!  
[dramirez@10 ~]$
```

```
dramirez@10:~  
Verificando : sysprof-capture-devel-3.40.1-3.el9.x86_64 18/18  
  
Instalado:  
brotli-1.0.9-9.el9_7.x86_64  
brotli-devel-1.0.9-9.el9_7.x86_64  
freetype-devel-2.10.4-10.el9_5.x86_64  
glib2-devel-2.68.4-18.el9_7.2.x86_64  
graphite2-devel-1.3.14-9.el9.x86_64  
harfbuzz-devel-2.7.4-10.el9.x86_64  
libblkid-devel-2.37.4-21.el9_7.x86_64  
libcurl-devel-7.76.1-35.el9_7.3.x86_64  
libffi-devel-3.4.2-8.el9.x86_64  
libicu-devel-67.1-10.el9_6.x86_64  
libjpeg-turbo-devel-2.0.90-7.el9.x86_64  
libmount-devel-2.37.4-21.el9_7.x86_64  
libpng-devel-2:1.6.37-12.el9_7.4.x86_64  
libselinux-devel-3.6-3.el9.x86_64  
libsepol-devel-3.6-3.el9.x86_64  
pcre2-devel-10.40-6.el9.x86_64  
pcre2-utf16-10.40-6.el9.x86_64  
sysprof-capture-devel-3.40.1-3.el9.x86_64  
  
¡Listo!  
[dramirez@10 ~]$
```

```
dramirez@10:~
Tamaño instalado: 736 k
Descargando paquetes:
(1/2): oniguruma-devel-6.9.6-1.el9.6.x86_64.rpm 35 kB/s | 65 kB 00:01
(2/2): systemd-devel-252-55.el9_7.9.alma.1.x86_ 214 kB/s | 481 kB 00:02
-----
Total 163 kB/s | 546 kB 00:03
Ejecutando verificación de operación
Verificación de operación exitosa.
Ejecutando prueba de operaciones
Prueba de operación exitosa.
Ejecutando operación
  Preparando      : 1/1
  Instalando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 1/2
  Instalando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Ejecutando scriptlet: systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Verificando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 1/2
  Verificando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 2/2

Instalado:
  oniguruma-devel-6.9.6-1.el9.6.x86_64
  systemd-devel-252-55.el9_7.9.alma.1.x86_64

¡Listo!
[dramirez@10 ~]$
```

CREACIÓN DE USUARIO Y GRUPO:

```
dramirez@10:~
(1/2): oniguruma-devel-6.9.6-1.el9.6.x86_64.rpm 35 kB/s | 65 kB 00:01
(2/2): systemd-devel-252-55.el9_7.9.alma.1.x86_ 214 kB/s | 481 kB 00:02
-----
Total 163 kB/s | 546 kB 00:03
Ejecutando verificación de operación
Verificación de operación exitosa.
Ejecutando prueba de operaciones
Prueba de operación exitosa.
Ejecutando operación
  Preparando      : 1/1
  Instalando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 1/2
  Instalando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Ejecutando scriptlet: systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Verificando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 1/2
  Verificando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 2/2

Instalado:
  oniguruma-devel-6.9.6-1.el9.6.x86_64
  systemd-devel-252-55.el9_7.9.alma.1.x86_64

¡Listo!
[dramirez@10 ~]$ sudo groupadd nginx
[dramirez@10 ~]$ sudo useradd -r -g nginx -s /sbin/nologin nginx
[dramirez@10 ~]$
```

DESCARGA Y COMPILACION DE NGINX

```
dramirez@10:/usr/local/src
(2/2): systemd-devel-252-55.el9_7.9.alma.1.x86_ 214 kB/s | 481 kB 00:02
-----
Total 163 kB/s | 546 kB 00:03
Ejecutando verificación de operación
Verificación de operación exitosa.
Ejecutando prueba de operaciones
Prueba de operación exitosa.
Ejecutando operación
  Preparando      : 1/1
  Instalando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 1/2
  Instalando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Ejecutando scriptlet: systemd-devel-252-55.el9_7.9.alma.1.x86_64 2/2
  Verificando      : systemd-devel-252-55.el9_7.9.alma.1.x86_64 1/2
  Verificando      : oniguruma-devel-6.9.6-1.el9.6.x86_64 2/2

Instalado:
  oniguruma-devel-6.9.6-1.el9.6.x86_64
  systemd-devel-252-55.el9_7.9.alma.1.x86_64

¡Listo!
[dramirez@10 ~]$ sudo groupadd nginx
[dramirez@10 ~]$ sudo useradd -r -g nginx -s /sbin/nologin nginx
[dramirez@10 ~]$ cd /usr/local/src
[dramirez@10 src]$ sudo wget https://nginx.org/download/nginx-1.31.0.tar.gz
```

```
dramirez@10:/usr/local/src
Instalado:
  oniguruma-devel-6.9.6-1.el9.6.x86_64
  systemd-devel-252-55.el9_7.9.alma.1.x86_64

¡Listo!
[dramirez@10 ~]$ sudo groupadd nginx
[dramirez@10 ~]$ sudo useradd -r -g nginx -s /sbin/nologin nginx
[dramirez@10 ~]$ cd /usr/local/src
[dramirez@10 src]$ sudo wget https://nginx.org/download/nginx-1.31.0.tar.gz
--2026-05-24 15:30:05-- https://nginx.org/download/nginx-1.31.0.tar.gz
Resolviendo nginx.org (nginx.org)... 3.147.99.198, 2a05:d014:5c0:2601::6, 2a05:d014:5c0:2600::6
Conectando con nginx.org (nginx.org)[3.147.99.198]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 1337335 (1.3M) [application/octet-stream]
Grabando a: «nginx-1.31.0.tar.gz»

nginx-1.31.0.tar.gz 100%[=====>] 1.27M 331KB/s en 4.5s

2026-05-24 15:30:11 (291 KB/s) - «nginx-1.31.0.tar.gz» guardado [1337335/1337335]

[dramirez@10 src]$ sudo tar -xzf nginx-1.31.0.tar.gz
```



```
dramirez@10:/usr/local/src/nginx-1.31.0
resolviendo nginx.org (nginx.org)... 3.147.99.198, 2a05:d014:5c0:2601::6, 2a05:d014:5c0:2600::6
Conectando con nginx.org (nginx.org)[3.147.99.198]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 1337335 (1.3M) [application/octet-stream]
Grabando a: «nginx-1.31.0.tar.gz»

nginx-1.31.0.tar.gz 100%[=====] 1.27M 331KB/s en 4.5s

2026-05-24 15:30:11 (291 KB/s) - «nginx-1.31.0.tar.gz» guardado [1337335/1337335]

[dramirez@10 src]$ sudo tar -xzf nginx-1.31.0.tar.gz
[dramirez@10 src]$ cd nginx-1.31.0
[dramirez@10 nginx-1.31.0]$ sudo ./configure \
> --prefix=/srv/nginx \
> --user=nginx \
> --group=nginx \
> --with-http_ssl_module
checking for OS
+ Linux 5.14.0-611.5.1.el9_7.x86_64 x86_64
checking for C compiler ... found
+ using GNU C compiler
+ gcc version: 11.5.0 20240719 (Red Hat 11.5.0-11) (GCC)
checking for openssl ... found
```

```
dramirez@10:/usr/local/src/nginx-1.31.0
checking for OpenSSL library ... found
checking for zlib library ... found
creating objs/Makefile

Configuration summary
+ using system PCRE2 library
+ using system OpenSSL library
+ using system zlib library

nginx path prefix: "/srv/nginx"
nginx binary file: "/srv/nginx/sbin/nginx"
nginx modules path: "/srv/nginx/modules"
nginx configuration prefix: "/srv/nginx/conf"
nginx configuration file: "/srv/nginx/conf/nginx.conf"
nginx pid file: "/srv/nginx/logs/nginx.pid"
nginx error log file: "/srv/nginx/logs/error.log"
nginx http access log file: "/srv/nginx/logs/access.log"
nginx http client request body temporary files: "client_body_temp"
nginx http proxy temporary files: "proxy_temp"
nginx http fastcgi temporary files: "fastcgi_temp"
nginx http uwsgi temporary files: "uwsgi_temp"
nginx http scgi temporary files: "scgi_temp"

[dramirez@10 nginx-1.31.0]$
```

ESCRIBIMOS SUDO MAKE Y SUDO MAKE INSTALL

```
dramirez@10:/usr/local/src/nginx-1.31.0
test -f '/srv/nginx/conf/fastcgi.conf' \
    || cp conf/fastcgi.conf '/srv/nginx/conf'
cp conf/fastcgi.conf '/srv/nginx/conf/fastcgi.conf.default'
test -f '/srv/nginx/conf/uwsgi_params' \
    || cp conf/uwsgi_params '/srv/nginx/conf'
cp conf/uwsgi_params \
    '/srv/nginx/conf/uwsgi_params.default'
test -f '/srv/nginx/conf/scgi_params' \
    || cp conf/scgi_params '/srv/nginx/conf'
cp conf/scgi_params \
    '/srv/nginx/conf/scgi_params.default'
test -f '/srv/nginx/conf/nginx.conf' \
    || cp conf/nginx.conf '/srv/nginx/conf/nginx.conf'
cp conf/nginx.conf '/srv/nginx/conf/nginx.conf.default'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
test -d '/srv/nginx/html' \
    || cp -R html '/srv/nginx'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
make[1]: se sale del directorio '/usr/local/src/nginx-1.31.0'
[dramirez@10 nginx-1.31.0]$
```

```
dramirez@10:/usr/local/src/nginx-1.31.0
    || cp conf/uwsgi_params '/srv/nginx/conf'
cp conf/uwsgi_params \
    '/srv/nginx/conf/uwsgi_params.default'
test -f '/srv/nginx/conf/scgi_params' \
    || cp conf/scgi_params '/srv/nginx/conf'
cp conf/scgi_params \
    '/srv/nginx/conf/scgi_params.default'
test -f '/srv/nginx/conf/nginx.conf' \
    || cp conf/nginx.conf '/srv/nginx/conf/nginx.conf'
cp conf/nginx.conf '/srv/nginx/conf/nginx.conf.default'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
test -d '/srv/nginx/html' \
    || cp -R html '/srv/nginx'
test -d '/srv/nginx/logs' \
    || mkdir -p '/srv/nginx/logs'
make[1]: se sale del directorio '/usr/local/src/nginx-1.31.0'
[dramirez@10 nginx-1.31.0]$ sudo nano /etc/systemd/system/nginx.service
[dramirez@10 nginx-1.31.0]$ /srv/nginx/conf/nginx.conf
bash: /srv/nginx/conf/nginx.conf: Permiso denegado
[dramirez@10 nginx-1.31.0]$ sudo nano /srv/nginx/conf/nginx.conf
```

```
dramirez@10:/usr/local/src/nginx-1.31.0 — sudo nano /etc/syst...
GNU nano 5.6.1 /etc/systemd/system/nginx.service
[Unit]
Description=NGINX Web Server
After=network.target

[Service]
Type=forking
PIDFile=/srv/nginx/logs/nginx.pid
ExecStartPre=/srv/nginx/sbin/nginx -t
ExecStart=/srv/nginx/sbin/nginx
ExecStop=/srv/nginx/sbin/nginx -s quit

[Install]
WantedBy=multi-user.target

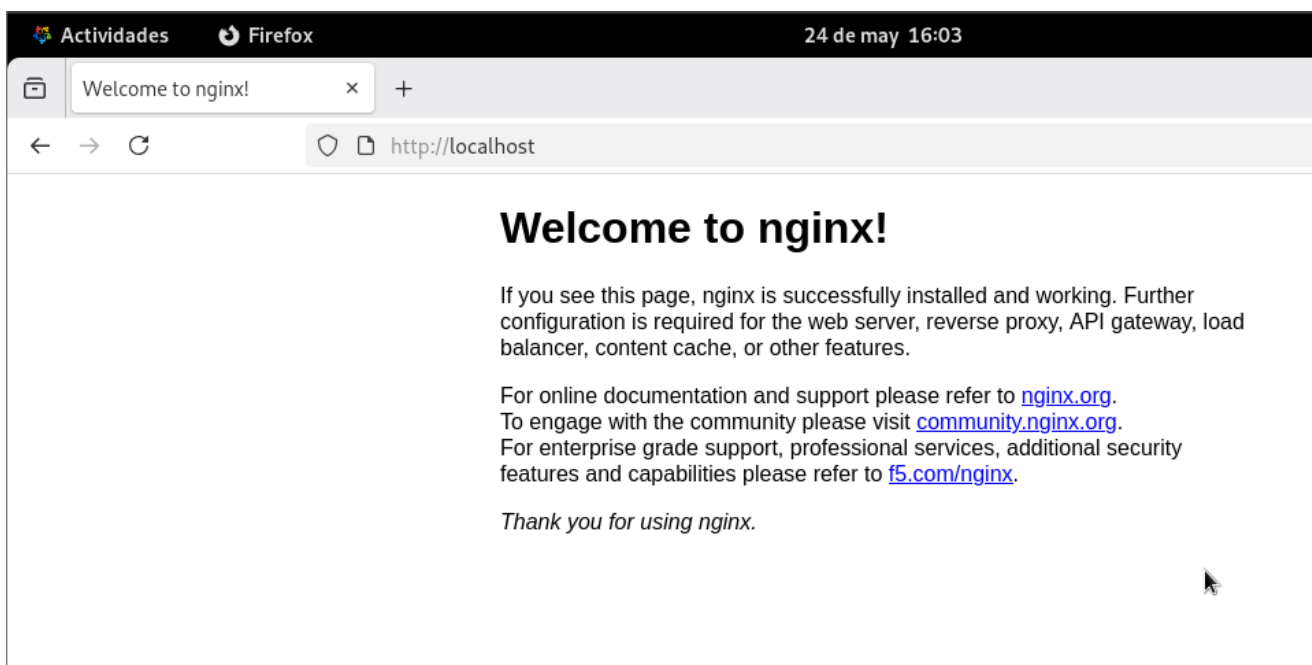
[ 13 líneas leídas ]
^G Ayuda ^O Guardar ^W Buscar ^K Cortar ^T Ejecutar ^C Ubicación
^X Salir ^R Leer fich. ^\ Reemplazar ^U Pegar ^J Justificar ^_ Ir a línea
```

HABILITACION DEL SERVICIO:

sudo systemctl daemon-reload

sudo systemctl enable nginx

sudo systemctl start nginx



DESCARGA Y COMPILACIÓN DE PHP

```
[dramirez@10 nginx-1.31.0]$ sudo groupadd php
[dramirez@10 nginx-1.31.0]$ sudo useradd -r -g nginx -s /sbin/nologin php
[dramirez@10 nginx-1.31.0]$
```

 dramirez@10:/usr/local/src

```
valid_lft forever preferred_lft forever
[dramirez@10 nginx-1.31.0]$ sudo systemctl restart nginx
[dramirez@10 nginx-1.31.0]$ sudo group add php
[sudo] password for dramirez:
sudo: group: command not found
[dramirez@10 nginx-1.31.0]$ sudo groupadd php
[dramirez@10 nginx-1.31.0]$ sudo useradd -r -g nginx -s /sbin/nologin php
[dramirez@10 nginx-1.31.0]$ cd /usr/local/src
[dramirez@10 src]$ sudo wget https://www.php.net/distributions/php-8.4.0.tar.g
--2026-05-24 16:18:18-- https://www.php.net/distributions/php-8.4.0.tar.gz
Resolviendo www.php.net (www.php.net)... 185.85.0.29, 2a02:cb40:200::1ad
Conectando con www.php.net (www.php.net)[185.85.0.29]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: no especificado [application/x-gzip]
Grabando a: «php-8.4.0.tar.gz»

php-8.4.0.tar.gz [ <=> ]

2026-05-24 16:18:33 (1.48 MB/s) - «php-8.4.0.tar.gz» guardado [21615912]

[dramirez@10 src]$
```

 dramirez@10:/usr/local/src/php-8.4.0   

```
[sudo] password for dramirez:
sudo: group: command not found
[dramirez@10 nginx-1.31.0]$ sudo groupadd php
[dramirez@10 nginx-1.31.0]$ sudo useradd -r -g nginx -s /sbin/nologin php
[dramirez@10 nginx-1.31.0]$ cd /usr/local/src
[dramirez@10 src]$ sudo wget https://www.php.net/distributions/php-8.4.0.tar.gz
--2026-05-24 16:18:18-- https://www.php.net/distributions/php-8.4.0.tar.gz
Resolviendo www.php.net (www.php.net)... 185.85.0.29, 2a02:cb40:200::1ad
Conectando con www.php.net (www.php.net)[185.85.0.29]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: no especificado [application/x-gzip]
Grabando a: «php-8.4.0.tar.gz»

php-8.4.0.tar.gz [ <=> ] 20.61M 4.45MB/s en 14s

2026-05-24 16:18:33 (1.48 MB/s) - «php-8.4.0.tar.gz» guardado [21615912]

[dramirez@10 src]$ sudo tar -xzf php-8.4.0.tar.gz
[dramirez@10 src]$ cd php-8.4.0
[dramirez@10 php-8.4.0]$ sudo ./configure --prefix=/srv/nginx --enable-fpm --with-fpm-user=php --with-fpm-group=
nginx --with-zlib --with-curl --with-openssl --enable-mbstring --enable-intl --with-jpeg --with-freetype
```

```
dramirez@10:/usr/local/src/php-8.4.0

config.status: creating scripts/php-config
config.status: creating scripts/phpize
config.status: creating main/php_config.h
config.status: executing Zend/zend_config.h commands
config.status: executing main/internal_functions.c commands
config.status: executing main/internal_functions_cli.c commands
config.status: executing default commands

+-----+
| License:
| This software is subject to the PHP License, available in this
| distribution in the file LICENSE. By continuing this installation
| process, you are bound by the terms of this license agreement.
| If you do not agree with the terms of this license, you must abort
| the installation process at this point.
+-----+

Thank you for using PHP.

configure: WARNING: unrecognized options: --with-fpm-goup
[dramirez@10 php-8.4.0]$
```

```
dramirez@10:/usr/local/src/php-8.4.0

[sudo] password for dramirez:
Installing shared extensions:      /srv/nginx/lib/php/extensions/no-debug-non-zts-20240924/
Installing PHP CLI binary:        /srv/nginx/bin/
Installing PHP CLI man page:      /srv/nginx/php/man/man1/
Installing PHP FPM binary:        /srv/nginx/sbin/
Installing PHP FPM defconfig:     /srv/nginx/etc/
Installing PHP FPM man page:      /srv/nginx/php/man/man8/
Installing PHP FPM status page:   /srv/nginx/php/php/fpm/
Installing phpdbg binary:         /srv/nginx/bin/
Installing phpdbg man page:       /srv/nginx/php/man/man1/
Installing PHP CGI binary:        /srv/nginx/bin/
Installing PHP CGI man page:      /srv/nginx/php/man/man1/
Installing build environment:     /srv/nginx/lib/php/build/
Installing header files:          /srv/nginx/include/php/
Installing helper programs:       /srv/nginx/bin/
    program: phpize
    program: php-config
Installing man pages:             /srv/nginx/php/man/man1/
    page: phpize.1
    page: php-config.1
[dramirez@10 php-8.4.0]$
```

CONFIGURACIÓN PHP-FPM

```
[dramirez@10 php-8.4.0]$ sudo cp /srv/nginx/etc/php-fpm.conf.default /srv/nginx/etc/php-fpm.conf
[dramirez@10 php-8.4.0]$ sudo mkdir -p /srv/nginx/etc/php-fpm.d
[dramirez@10 php-8.4.0]$ sudo cp /usr/local/src/php-8.4.0/sapi/fpm/www.conf /srv/nginx/etc/php-fpm.d/www.conf
[dramirez@10 php-8.4.0]$
```

CREAMOS SERVICIO

```
dramirez@10:/usr/local/src/php-8.4.0 — sudo nano /etc/systemd/system/php-fpm8.4.service
GNU nano 5.6.1 /etc/systemd/system/php-fpm8.4.service Modificado

[Unit]
Description=PHP 8.4 FastCGI Process Manager
After=network.target

[Service]
Type=forking
PIDFile=/srv/nginx/var/run/php-fpm.pid
ExecStart=/srv/nginx/sbin/php-fpm
ExecReload=/bin/kill -USR2 $MAINPID

[Install]
WantedBy=multi-user.target

^G Ayuda ^O Guardar ^W Buscar ^K Cortar ^T Ejecutar ^C Ubicación M-U Deshacer
^X Salir ^R Leer fich. ^\ Reemplazar ^U Pegar ^J Justificar ^_ Ir a línea M-E Rehacer
```

```
dramirez@10:/usr/local/src/php-8.4.0
Installing build environment: /srv/nginx/lib/php/build/
Installing header files: /srv/nginx/include/php/
Installing helper programs: /srv/nginx/bin/
  program: phpize
  program: php-config
Installing man pages: /srv/nginx/php/man/man1/
  page: phpize.1
  page: php-config.1
[dramirez@10 php-8.4.0]$ sudo cp /srv/nginx/etc/php-fpm.conf.default /srv/nginx/etc/php-fpm.conf
[dramirez@10 php-8.4.0]$ sudo mkdir -p /srv/nginx/etc/php-fpm.d
[dramirez@10 php-8.4.0]$ sudo cp /usr/local/src/php-8.4.0/sapi/fpm/www.conf /srv/nginx/etc/php-fpm.d/www.conf
[dramirez@10 php-8.4.0]$ sudo nano /srv/nginx/etc/php-fpm.d/www.conf
[dramirez@10 php-8.4.0]$ sudo nano /srv/nginx/etc/php-fpm.d/www.conf
[dramirez@10 php-8.4.0]$ sudo nano /etc/systemd/system/php-fpm8.4.service
[dramirez@10 php-8.4.0]$ /srv/nginx/etc/php-fpm.d/www.conf
bash: /srv/nginx/etc/php-fpm.d/www.conf: No existe el fichero o el directorio
[dramirez@10 php-8.4.0]$ /srv/nginx/etc/php-fpm.d/www.conf
bash: /srv/nginx/etc/php-fpm.d/www.conf: Permiso denegado
[dramirez@10 php-8.4.0]$ sudo systemctl daemon-reload
[sudo] password for dramirez:
[dramirez@10 php-8.4.0]$
```

```
[dramirez@10 php-8.4.0]$ sudo nano /etc/systemd/system/php-fpm8.4.service
[dramirez@10 php-8.4.0]$ sudo systemctl daemon-reload
[dramirez@10 php-8.4.0]$ sudo systemctl restart php-fpm8.4
[dramirez@10 php-8.4.0]$ sudo systemctl status php-fpm8.4
● php-fpm8.4.service - PHP 8.4 FastCGI Process Manager
   Loaded: loaded (/etc/systemd/system/php-fpm8.4.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-24 17:08:17 CST; 9s ago
     Process: 293966 ExecStart=/srv/nginx/sbin/php-fpm (code=exited, status=0/SUCCESS)
    Main PID: 293968 (php-fpm)
      Tasks: 3 (limit: 22972)
     Memory: 7.8M (peak: 10.2M)
        CPU: 40ms
    CGroup: /system.slice/php-fpm8.4.service
            └─293968 "php-fpm: master process (/srv/nginx/etc/php-fpm.conf)"
               └─293969 "php-fpm: pool www"
                  └─293970 "php-fpm: pool www"

may 24 17:08:17 10.0.2.15 systemd[1]: Starting PHP 8.4 FastCGI Process Manager...
may 24 17:08:17 10.0.2.15 systemd[1]: Started PHP 8.4 FastCGI Process Manager.
[dramirez@10 php-8.4.0]$
```

CONECTAR NGINX CON PHP

```
dramirez@10:/usr/local/src/php-8.4.0 — sudo nano /srv/nginx/conf/nginx.conf
GNU nano 5.6.1 /srv/nginx/conf/nginx.conf
server {
    listen      80;
    server_name localhost;

    #access_log logs/host.access.log main;

    location / {
        root    html;
        index   index.html index.htm;
    }
    location ~ \.php$ {
        root    html;
        fastcgi_pass unix:/tmp/php84.sock;
        fastcgi_index index.php;
        include fastcgi.conf;
    }
    #error_page 404              /404.html;
}
```

^G Ayuda ^O Guardar ^W Buscar ^K Cortar ^T Ejecutar ^C Ubicación M-U Deshacer
^X Salir ^R Leer fich. ^\ Reemplazar ^U Pegar ^J Justificar ^_ Ir a línea M-E Rehacer

CREAR ARCHIVO PHP

```
dramirez@10:/usr/local/src/php-8.4.0
Loaded: loaded (/etc/systemd/system/nginx.service; enabled; preset: disabled)
Active: active (running) since Sun 2026-05-24 17:20:13 CST; 1min 50s ago
Process: 294149 ExecStartPre=/srv/nginx/sbin/nginx -t (code=exited, status=0/SUCCESS)
Process: 294151 ExecStart=/srv/nginx/sbin/nginx (code=exited, status=0/SUCCESS)
Main PID: 294152 (nginx)
Tasks: 2 (limit: 22972)
Memory: 2.1M (peak: 2.2M)
CPU: 16ms
CGroup: /system.slice/nginx.service
├─294152 "nginx: master process /srv/nginx/sbin/nginx"
└─294153 "nginx: worker process"

may 24 17:20:13 10.0.2.15 systemd[1]: Starting NGINX Web Server...
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: the configuration file /srv/nginx/conf/nginx.conf syntax is ok
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: configuration file /srv/nginx/conf/nginx.conf test is successful
may 24 17:20:13 10.0.2.15 systemd[1]: nginx.service: Failed to parse PID from file /srv/nginx/logs/nginx.pid: >
may 24 17:20:13 10.0.2.15 systemd[1]: Started NGINX Web Server.
~
~
[dramirez@10 php-8.4.0]$ sudo nano /srv/nginx/html/phpinfo.php
```



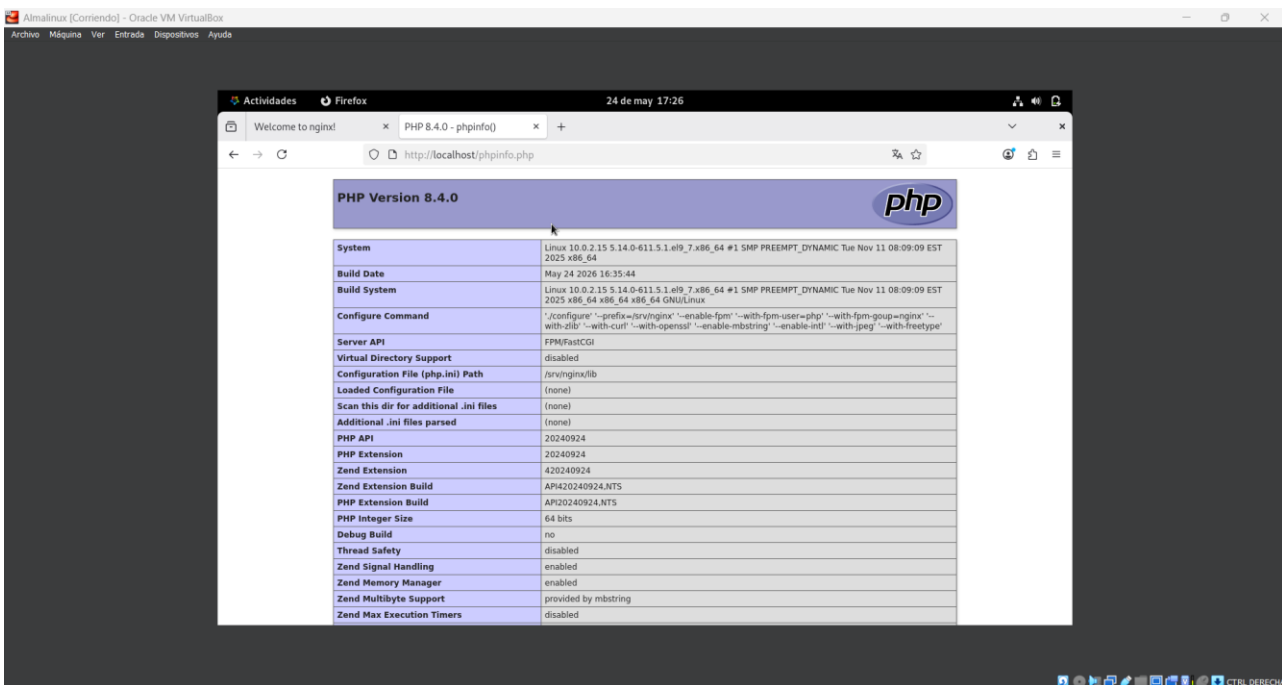
```
dramirez@10:/usr/local/src/php-8.4.0 — sudo nano /srv/nginx/html/phpinfo.php
GNU nano 5.6.1 /srv/nginx/html/phpinfo.php
?php
phpinfo();
?>
```

[3 líneas leídas]

^G Ayuda ^O Guardar ^W Buscar ^K Cortar ^T Ejecutar ^C Ubicación M-U Deshacer
^X Salir ^R Leer fich. ^\ Reemplazar ^U Pegar ^J Justificar ^_ Ir a línea M-E Rehacer

ENTRAMOS A:

<http://localhost/phpinfo.php>



VERIFICAMOS AUTOARRANQUE:

Estatus de nginx:

```
[dramirez@10 php-8.4.0]$ sudo systemctl status nginx
● nginx.service - NGINX Web Server
   Loaded: loaded (/etc/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-24 17:20:13 CST; 12min ago
     Main PID: 294152 (nginx)
        Tasks: 2 (limit: 22972)
       Memory: 2.2M (peak: 2.7M)
          CPU: 17ms
      CGroup: /system.slice/nginx.service
              └─294152 "nginx: master process /srv/nginx/sbin/nginx"
                 └─294153 "nginx: worker process"

may 24 17:20:13 10.0.2.15 systemd[1]: Starting NGINX Web Server...
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: the configuration file /srv/nginx/conf/nginx.conf syntax is ok
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: configuration file /srv/nginx/conf/nginx.conf test is successful
may 24 17:20:13 10.0.2.15 systemd[1]: nginx.service: Failed to parse PID from file /srv/nginx/logs/nginx.pid:
may 24 17:20:13 10.0.2.15 systemd[1]: Started NGINX Web Server.
lines 1-16/16 (END)
```

Estatus de php:

```
[dramirez@10 php-8.4.0]$ sudo systemctl status php-fpm8.4
● php-fpm8.4.service - PHP 8.4 FastCGI Process Manager
   Loaded: loaded (/etc/systemd/system/php-fpm8.4.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-24 17:08:17 CST; 24min ago
     Main PID: 293968 (php-fpm)
        Tasks: 3 (limit: 22972)
       Memory: 9.4M (peak: 10.2M)
          CPU: 85ms
      CGroup: /system.slice/php-fpm8.4.service
              └─293968 "php-fpm: master process (/srv/nginx/etc/php-fpm.conf)"
                 └─293969 "php-fpm: pool www"
                    └─293970 "php-fpm: pool www"

may 24 17:08:17 10.0.2.15 systemd[1]: Starting PHP 8.4 FastCGI Process Manager...
may 24 17:08:17 10.0.2.15 systemd[1]: Started PHP 8.4 FastCGI Process Manager.
[dramirez@10 php-8.4.0]$
```

Version de nginx:

```
may 24 17:08:17 10.0.2.15 systemd[1]: Started PHP 8.4 FastCGI Process Manager.
[dramirez@10 php-8.4.0]$ /srv/nginx/sbin/nginx -v
nginx version: nginx/1.31.0
[dramirez@10 php-8.4.0]$
```

Estado de NGINX:

```
[dramirez@10 php-8.4.0]$ sudo systemctl status nginx
● nginx.service - NGINX Web Server
   Loaded: loaded (/etc/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-24 17:20:13 CST; 12min ago
     Main PID: 294152 (nginx)
        Tasks: 2 (limit: 22972)
       Memory: 2.2M (peak: 2.7M)
          CPU: 17ms
      CGroup: /system.slice/nginx.service
              └─294152 "nginx: master process /srv/nginx/sbin/nginx"
                 └─294153 "nginx: worker process"

may 24 17:20:13 10.0.2.15 systemd[1]: Starting NGINX Web Server...
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: the configuration file /srv/nginx/conf/nginx.conf syntax is ok
may 24 17:20:13 10.0.2.15 nginx[294149]: nginx: configuration file /srv/nginx/conf/nginx.conf test is successful
may 24 17:20:13 10.0.2.15 systemd[1]: nginx.service: Failed to parse PID from file /srv/nginx/logs/nginx.pid:
may 24 17:20:13 10.0.2.15 systemd[1]: Started NGINX Web Server.
lines 1-16/16 (END)
```

Version de PHP

```
[dramirez@10 php-8.4.0]$ /srv/nginx/bin/php -v
PHP 8.4.0 (cli) (built: May 24 2026 16:37:21) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.4.0, Copyright (c) Zend Technologies
[dramirez@10 php-8.4.0]$
```

Estado de php:

```
[dramirez@10 php-8.4.0]$ sudo systemctl status php-fpm8.4
● php-fpm8.4.service - PHP 8.4 FastCGI Process Manager
   Loaded: loaded (/etc/systemd/system/php-fpm8.4.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-24 17:08:17 CST; 24min ago
     Main PID: 293968 (php-fpm)
       Tasks: 3 (limit: 22972)
      Memory: 9.4M (peak: 10.2M)
         CPU: 85ms
    CGroup: /system.slice/php-fpm8.4.service
            └─293968 "php-fpm: master process (/srv/nginx/etc/php-fpm.conf)"
               └─293969 "php-fpm: pool www"
                  └─293970 "php-fpm: pool www"

may 24 17:08:17 10.0.2.15 systemd[1]: Starting PHP 8.4 FastCGI Process Manager...
may 24 17:08:17 10.0.2.15 systemd[1]: Started PHP 8.4 FastCGI Process Manager.
[dramirez@10 php-8.4.0]$
```

Socket funcionando:

```
[dramirez@10 php-8.4.0]$ ls -l /tmp/php84.sock
srw-rw----. 1 nginx nginx 0 may 24 17:08 /tmp/php84.sock
[dramirez@10 php-8.4.0]$
```

Servidores habilitados al arranque:

```
[dramirez@10 php-8.4.0]$ sudo systemctl is-enabled nginx
[sudo] password for dramirez:
enabled
[dramirez@10 php-8.4.0]$ sudo systemctl is-enabled php-fpm8.4
enabled
[dramirez@10 php-8.4.0]$
```

CONFIGURACIÓN FASTCGI EN NGINX:

```
GNU nano 5.6.1 /s
#user  nginx nginx;
worker_processes  1;

#error_log  logs/error.log;
#error_log  logs/error.log  notice;
#error_log  logs/error.log  info;

#pid        logs/nginx.pid;

events {
    worker_connections  1024;
}

http {
```

```
GNU nano 5.6.1 /srv/nginx
server {
    listen      80;
    server_name localhost;

    #access_log  logs/host.access.log  main;

    location / {
        root    html;
        index   index.html index.htm;
    }
    location ~ /\.php$ {
        root    html;
        fastcgi_pass  unix:/tmp/php84.sock;
        fastcgi_index index.php;
        include fastcgi.conf;
    }
}
```

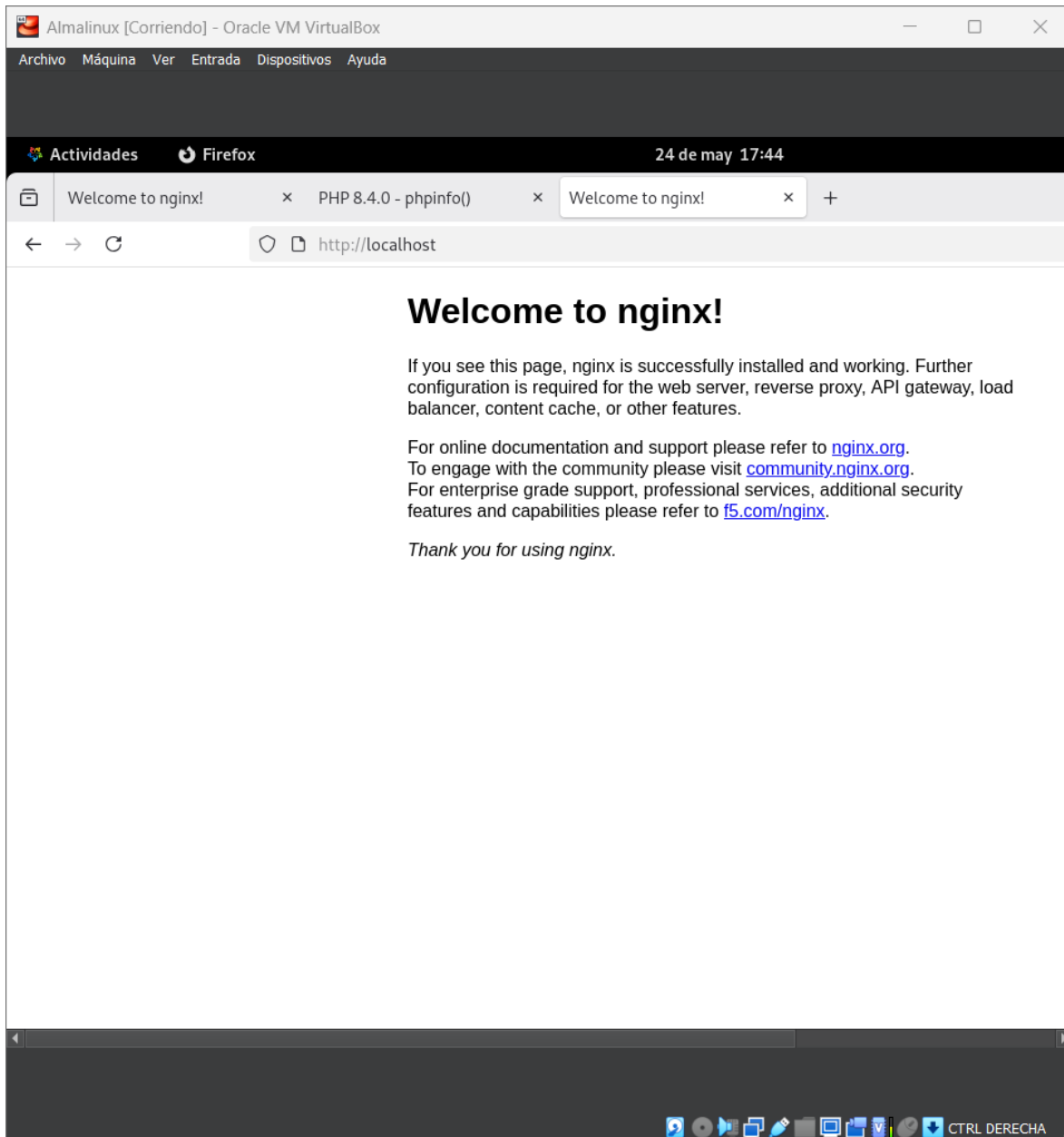
CONFIGURACIÓN PHP-FPM

```
GNU nano 5.6.1 /srv/nginx/etc/php-fpm.d/www.conf
; 'ip.address:port' - to listen on a TCP socket to a specific IPv4 address
;                   a specific port;
; '[ip6:address]:port' - to listen on a TCP socket to a specific IPv6 address
;                   a specific port;
; 'port' - to listen on a TCP socket to all addresses
;         (IPv6 and IPv4-mapped) on a specific port;
; '/path/to/unix/socket' - to listen on a unix socket.
; Note: This value is mandatory.
listen = /tmp/php84.sock

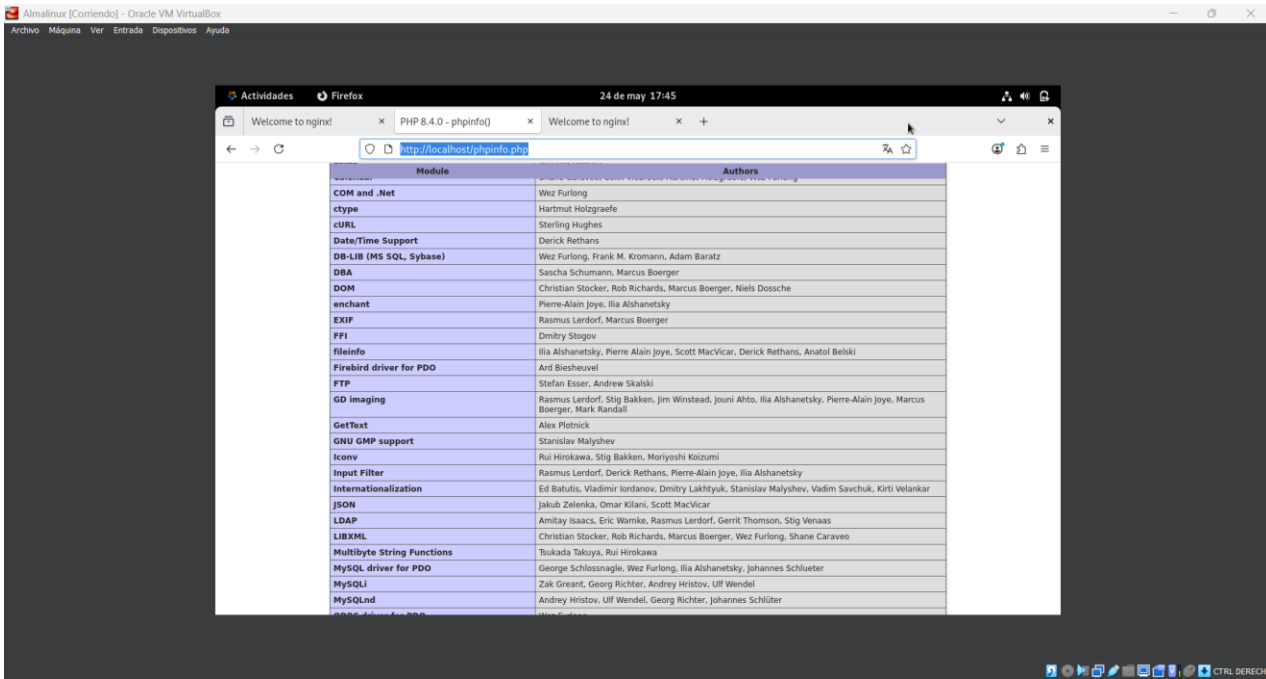
; Set listen(2) backlog.
; Default Value: 511 (-1 on Linux, FreeBSD and OpenBSD)
listen.backlog = 511

; Set permissions for unix socket, if one is used. In Linux, read/write
; permissions must be set in order to allow connections from a web server. Many
; BSD-derived systems allow connections regardless of permissions. The owner
```

PÁGINA PRINCIPAL NGINX FUNCIONANDO



PHP-FPM :



ESTRUCTURA INSTALADA EN /SRV/NGINX:

```
[dramirez@10 php-8.4.0]$ ls /srv/nginx
bin          conf  fastcgi_temp  include  logs  proxy_temp  scgi_temp  var
client_body_temp  etc  html          lib       php   sbin        uwsgi_temp
```

SERVICIO SYSTEMD CREADO

```
[dramirez@10 php-8.4.0]$ ls /etc/systemd/system | grep nginx
nginx.service
[dramirez@10 php-8.4.0]$ ls /etc/systemd/system | grep php
php-fpm8.4.service
[dramirez@10 php-8.4.0]$
```

CONCLUSION

Durante el desarrollo del proyecto se logró implementar correctamente un servidor NGINX compilado desde código fuente junto con PHP-FPM utilizando comunicación FastCGI mediante socket UNIX. También se configuraron servicios administrados mediante SystemD para garantizar el arranque automático del servidor y se verificó el correcto funcionamiento mediante pruebas realizadas con el archivo phpinfo.php. Este proyecto permitió comprender el proceso de compilación manual de servicios web, la administración de procesos y la integración de tecnologías en sistemas Linux.

REFERENCIAS

NGINX. (2026). NGINX Documentation. <https://nginx.org/>

PHP Group. (2026). PHP Documentation. <https://www.php.net/docs.php>

AlmaLinux OS Foundation. (2026). AlmaLinux Documentation. <https://wiki.almalinux.org/>